

| $x$ | sigmoid | tanh   | ReLU | LR |
|-----|---------|--------|------|----|
| 5   | 0.9933  | 0.999  | 5    | 5  |
| 7   | 0.999   | 0.999  | 7    | 7  |
| -1  | 0.268   | -0.761 | 0    | 0  |



22/08/25

exp: 5

## 5. STUDY OF ACTIVATION FUNCTIONS AND ITS ROLE

Aim: To study the activation functions and its role

Objective:

\* To understand why activation functions are necessary in neural network

\* To study commonly used activation fn: Sigmoid, Tanh, Relu, leaky Relu, Softmax

\* To compare their advantages disadvantages and suitable application

Pseudocode

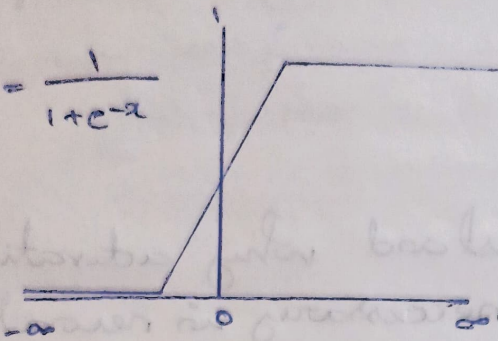
- 1) Import required libraries, l
- 2) Define mathematical functions for sigmoid, tanh, relu, leaky relu, softmax
3. Generate a range of input values
4. Compute outputs of all activation functions
5. plot ~~graph~~ of each activation functions
6. Compare their behaviour  
note observation



# Activation Function

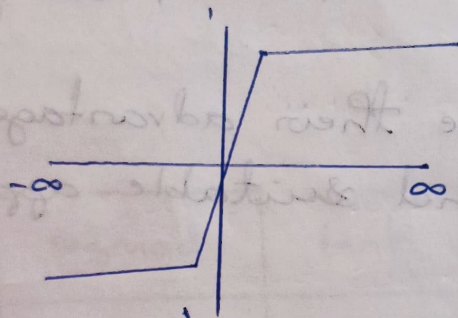
1. Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



2. tanh

$$\tanh(x)$$



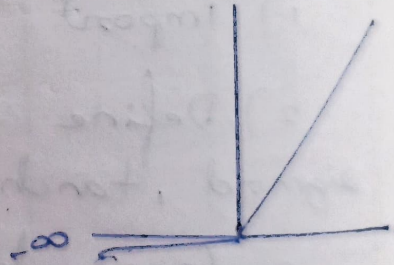
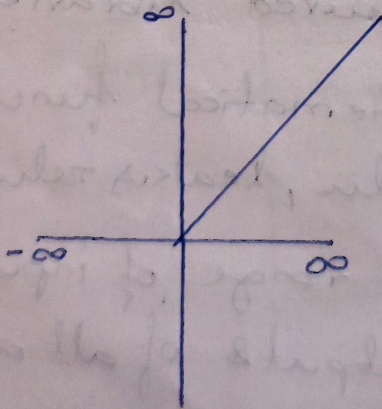
5. Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

6. ELU

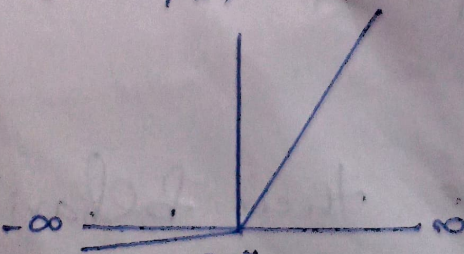
$$\begin{cases} x & x \geq 0 \\ \alpha(\sigma^x - 1) & x < 0 \end{cases}$$

3. Relu



4. leaky Relu

$$\max(0.1x, x)$$





## observation Table

| S <sub>No</sub> | Activation fn | output Range        | Advantage                              | Use case                                   |
|-----------------|---------------|---------------------|----------------------------------------|--------------------------------------------|
| 1               | Sigmoid       | $(0, 1)$            | Smooth probabilistic output            | Binary Classification                      |
| 2               | Tanh          | $(-1, 1)$           | Centered around 0; better than Sigmoid | Hidden layer<br>Convolutional network      |
| 3               | Relu          | $(0, \infty)$       | Fast, reduce computation Time          | Hidden layer<br>Convolutional (CNN)<br>ANN |
| 4               | leaky Relu    | $(-\infty, \infty)$ | Fines Relu (dying) issues              | Deep Hidden Layer                          |
| 5               | softmax       | $(0, 1)$ , sum=1    | Gives probability distribution         | output layer for multi-class               |

\* Relu and leaky Relu are most softmax in hidden layer

\* Sigmoid and Tanh are rarely used today due to vanishing gradient

\* Softmax for multiclass classification problems

### Result

Studied different activation function of their roles.

19/9/25